

Notes Week 9

a. Validating Tides Implementation

In the output, the convective turnover timescale was not changing. I traced this bug down to the following line of code:

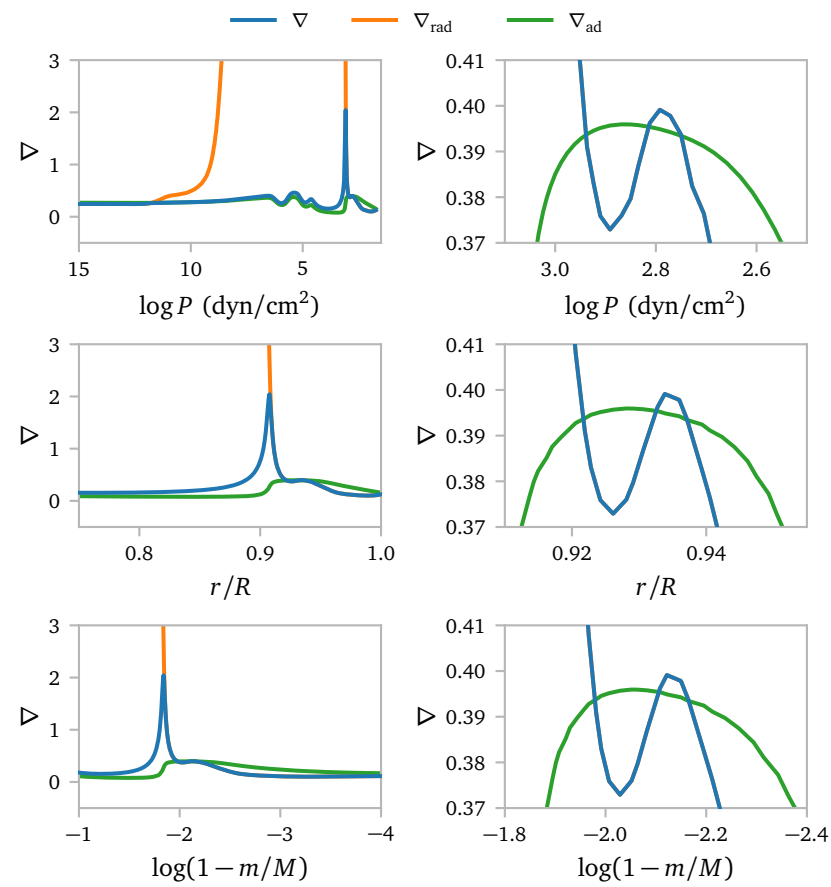
```
t_conv = 0.4311 * (3 * M_conv * R_conv**2 / (s% photosphere_L))**(1/3) * secyer
```

Here, the integer division in the exponent leads to $x^0 = 1$ always. Another problem revealed itself when analysing the output.

```
R_top = 380.72570878310302
R_bot = 365.63276071638143
R_conv= 15.092948066721579
in compute_t_conv: M_conv = 1.7530753171383301E-003
in compute_t_conv: R_conv = 15.092948066721579
in compute_t_conv: L = 8637.1120791303983
5.1764712937202029E-002
M_conv= 1.7530753171383301E-003 Msun
R_conv= 15.092948066721579 Rsun
t_conv= 704232.07225991588 s
t_conv= 0.26778921296673353 months
k_over_T = 7.7600225066267609E-013 /s
Omega_star = 2.6920249879859447E-008 /s
Omega_orb = 2.4471986527612926E-008 /s
J_orb = 3.5784679676889668E+053 g cm^2 / s
a = 91940611174289.297 cm
dt = 888553195.76110470 s
I_eff = 2.8898415232377608E+059 g cm^2
rg2 = 0.10857893011391405
```

The convective regions give wrong results, likely due to the superadiabicity of the outer layers looking at the minimal mass involved in the convective region.

Below, I plot the entropy profile of a star where my code finds a very tiny convective mass and radius:



In the superadiabatic region, there is a small bump, just large enough to create a very small convective region. The width of this bump aligns quite nicely with the values I find for the convective region.

Therefore, I should adjust the methods to find the mass and radius of the convective envelope region. My idea is to look for *all* convective regions that start sufficiently close to the outer edge of the star, and figure out which is the largest. I use this one for the final result.

```
real(dp) function compute_M_conv(s) result(M_conv)
implicit none

type(star_info), pointer :: s
integer :: i
real(dp) :: q_top, q_bot, q_max

M_conv = 0
q_max = 0
do i = 1, s% n_conv_regions
    if (abs(s% cz_top_mass(i) / Msun / s% star_mass - 1d0) < 3d-2) then
        ! --- starts sufficiently close to the edge
        q_top = s% cz_top_mass(i) / Msun / s% star_mass
        q_bot = s% cz_bot_mass(i) / Msun / s% star_mass

        write(*,*) q_top, q_bot

        if (q_top - q_bot > q_max) then
            q_max = q_top - q_bot
            write(*,*) "found new q_max:"
            write(*,*) q_top, q_bot, q_max
        end if
    end if
end do
write(*,*) "found M_conv:"
write(*,*) q_max, "gives M_conv = ", s% mstar * q_max / Msun

M_conv = s% mstar * q_max / Msun

end function compute_M_conv

real(dp) function compute_R_conv(s) result(R_conv)
implicit none

type(star_info), pointer :: s
real(dp) :: R_bot, R_top
real(dp) :: q_top, q_bot, q_max
real(dp) :: q_top_max, q_bot_max
integer :: k, i

R_conv = 0

! --- first find q_top and q_bot

q_max = 0
q_top_max = 0
q_bot_max = 0

do i = 1, s% n_conv_regions
    if (abs(s% cz_top_mass(i) / Msun / s% star_mass - 1d0) < 3d-2) then
        ! --- starts sufficiently close to the edge
        q_top = s% cz_top_mass(i) / Msun / s% star_mass
        q_bot = s% cz_bot_mass(i) / Msun / s% star_mass

        if (q_top - q_bot > q_max) then
            q_max = q_top - q_bot
            q_top_max = q_top
            q_bot_max = q_bot
        end if
    end if
end do

! --- now find which rs this corresponds to
R_top = 0d0

do k = 1, s% nz
    if (s% m(k) / Msun / s% star_mass <= q_top_max) then
        R_top = s% r(k)
        exit
    end if
end do

R_bot = 0d0 ! for safety

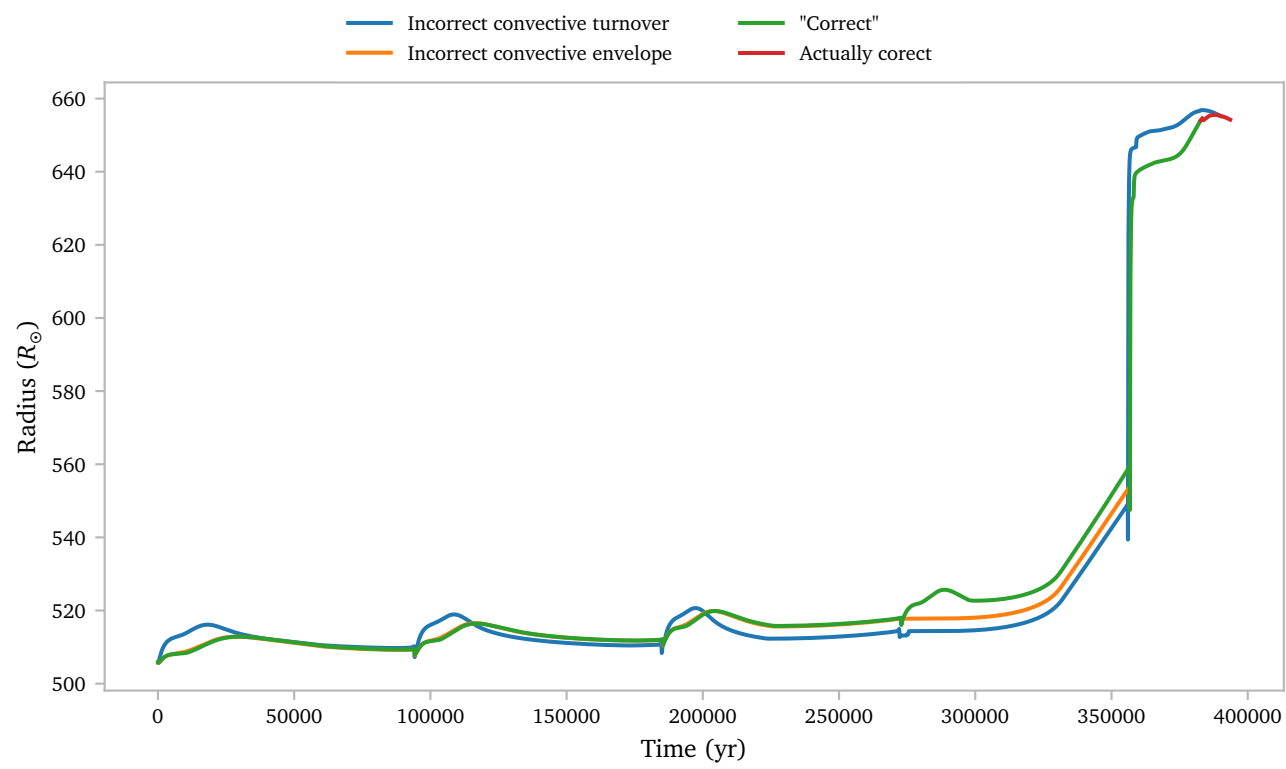
do k = k, s% nz
    if (s% m(k) / Msun / s% star_mass <= q_bot_max) then
        R_bot = s% r(k)
        exit
    end if
end do

R_conv = (R_top - R_bot) / Rsun

end function compute_R_conv
```

This results in the "correct" convective regions¹.

¹ At least, that is what I thought.



The "Correct" tide model terminates before the envelope is fully stripped. Again, my tides implementation cannot find the correct size of the convective envelope.

```
rl_relative_gap for each component
↳ NaN NaN NaN
rl for each component NaN
↳ NaN
r for each component 35119709581892.098
↳ 0.000000000000000000
m for each component 1.4776735316624046E+033
↳ 4.2199201770478900E+033
separation, angular momentum NaN
↳ NaN
jdot, jdot_mb, jdot_gr, jdot_ml:
↳ NaN 0.000000000000000000
↳ -1.2689479219736663E+024 -9.0373276853257355E+040
jdot_ls, jdot_missing_wind, extra_jdot:
↳ NaN 0.000000000000000000 0.000000000000000000
error solving rl_rel_gap
retry 21509
found M_conv:
0.000000000000000000 gives M_conv =
↳ 0.000000000000000000
M_conv= 0.000000000000000000 Msun
R_conv= 0.000000000000000000 Rsun
t_conv= 0.000000000000000000 s
t_conv= 0.000000000000000000 months
k_over_T = NaN /s
Omega_star = NaN /s
Omega_orb = 1.0050389612588993E-008 /s
J_orb = 2.6621375288468520E+053 g cm^2 / s
a = 155565622649979.09 cm
dt = 91137198.467318997 s
I_eff = 8.4459593589120531E+058 g cm^2
rg2 = 4.6331363281327982E-002
```

In my code, I do

```
do i = 1, s% n_conv_regions
  if (abs(s% cz_top_mass(i) / Msun / s% star_mass - 1d0) < 3d-2) then
    ! --- find envelope
```

to find envelope regions. This however, fails when the envelope becomes small, as relative to the star mass, the superadiabatic region grows and the convective envelope starts deeper into the star, surpassing $0.03M_{\text{star}}$. Really, we should just check the largest convective region that is NOT the core. For TPAGB-stars, the core is radiative, so we could probably just take the largest convective region. However, for the sake of generality, I still want to implement the core criterion. We can define the core mass using the mesa variable `s% he_core_mass`. Then, the new check is trivial:

```
do i = 1, s% n_conv_regions
  if (s% cz_top_mass(i) / Msun - s% he_core_mass > 8d-3) then
    ! --- find envelope
```

Here, I use `0.008` as a minimum mass above which the envelope should start. I think this is fine, since the simulation only goes down to $0.02M_{\odot}$.

This results in the red line from the plot above. The discontinuity arises from the sudden jump in convective envelope mass, but the important thing is that the model terminates due to a stripped envelope.

These are the typical values of the tides at the start of the evolution:

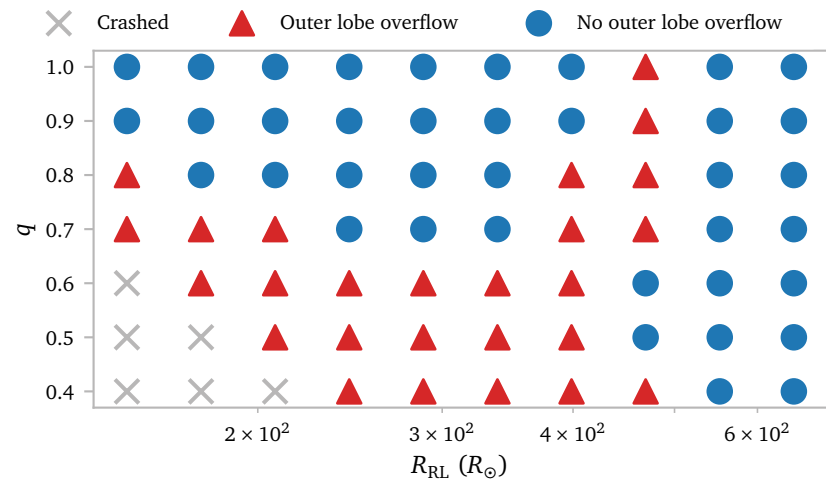
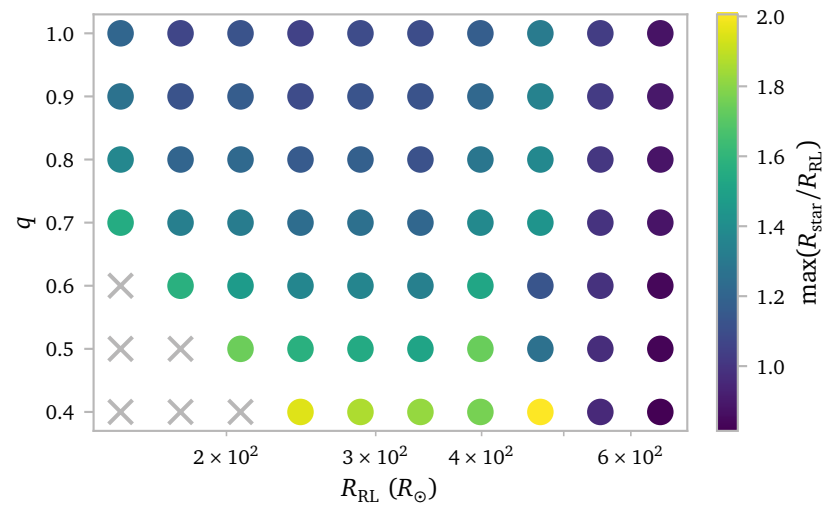
```
M_conv= 1.4104929944399043 Msun
R_conv= 346.63684044545323 Rsun
t_conv= 51634029.747670218 s
t_conv= 19.634204026036283 months
k_over_T = 2.1318031522939941E-009 /s
Omega_star = 2.4405051105719117E-008 /s
Omega_orb = 2.6326289142027669E-008 /s
J_orb = 3.6267648430630879E+053 g cm^2 / s
a = 88316477043967.516 cm
dt = 2524608000.0000000 s
I_eff = 2.7455090070529471E+059 g cm^2
rg2 = 0.11698324528595912
```

And these are typical values at the end of the evolution:

```
M_conv= 1.9333378562878496E-002 Msun
R_conv= 275.58270883774799 Rsun
t_conv= 11041685.311094698 s
t_conv= 4.1986787250341084 months
k_over_T = 2.8328298198011495E-010 /s
Omega_star = 1.7601613982742538E-008 /s
Omega_orb = 9.2259421941393170E-009 /s
J_orb = 2.3516310877365511E+053 g cm^2 / s
a = 162522837053206.97 cm
dt = 31346875.096955534 s
I_eff = 9.0310976468297850E+057 g cm^2
rg2 = 1.5792074851643058E-002
```

I think (?) these values look alright. The change in convective envelope mass makes sense due to the mass loss from RLOF and wind. With similar radii, this decreases the convective turnover timescale as well. Due to RLOF, the orbit expands drastically which decreases the orbital frequency and increases a . To me it also makes sense that the moment of inertia decreases, as is lost from the system while maintaining similar radii. This also describes why the gyration radius decreases.

b. Analytical Period, Mass-Ratio, Roche-Lobe Radius Relations



c. Investigating Curiosities TPAGB-grid

d. Wind Loss

MESA accounts for orbital angular momentum changes using the standard_jdot_ml routine. Below, I show the full thing:

```
subroutine default_jdot_ml(binary_id, ierr)
  integer, intent(in) :: binary_id
  integer, intent(out) :: ierr
  type (binary_info), pointer :: b
  real(dp) :: alfa
  ierr = 0
  call binary_ptr(binary_id, b, ierr)
  if (ierr /= 0) then
    write(*,*) 'failed in binary_ptr'
    return
  end if
  !mass lost from vicinity of donor
  b% jdot_ml = (b% mdot_system_transfer(b% d_i) + b% mdot_system_wind(b% d_i))*&
    pow2(b% m(b% a_i)/(b% m(b% a_i)+b% m(b% d_i))*b% separation)*2*pi/b% period *&
    sqrt(1 - pow2(b% eccentricity))
  !mass lost from vicinity of accretor
  b% jdot_ml = b% jdot_ml + (b% mdot_system_transfer(b% a_i) + b% mdot_system_wind(b% a_i))*&
    pow2(b% m(b% d_i)/(b% m(b% a_i)+b% m(b% d_i))*b% separation)*2*pi/b% period *&
    sqrt(1 - pow2(b% eccentricity))
  !mass lost from circumbinary coplanar toroid
  b% jdot_ml = b% jdot_ml + b% mdot_system_cct * b% mass_transfer_gamma * &
    sqrt(standard_cgrav * (b% m(1) + b% m(2)) * b% separation)
end subroutine default_jdot_ml
```

The change in orbital angular momentum is clearly the sum of three terms. First, a term for mass loss from the donor. This term is quite similar to the fast winds approximation or the Jeans mode (Jeans, 1924), which is generally described using the specific angular momentum as:

$$h = \frac{J}{M_d} = a_d^2 \omega = \left(\frac{M_a}{M_a + M_d} \right)^2 \sqrt{G(M_a + M_d)a}$$
$$= \left(\frac{M_a a}{M_a + M_d} \right)^2 \sqrt{G \frac{M_a + M_d}{a^3}}.$$

Now, applying Kepler's third law, this reduces to

$$h = \left(\frac{M_a a}{M_a + M_d} \right)^2 \frac{2\pi}{T}$$
$$\frac{dJ}{dt} = h \frac{dM_d}{dt} = (\dot{M}_{d,wind} + \dot{M}_{d,transfer}) \left(\frac{M_a a}{M_a + M_d} \right)^2 \frac{2\pi}{T}$$

In the absence of eccentricity, this is exactly the first equation in the MESA implementation. The second equation is simply the same, just now for the accretor.

The last equation is again just the Jeans mode, but now for material escaping from a circumbinary coplanar toroid². There are two new terms here though, $\gamma_{mass\ transfer}$ and $\dot{M}_{cct}$. The new term, $\gamma_{mass\ transfer}$ just determines the radius of the circumbinary coplanar toroid (cct), where $r_{cct} = \gamma_{mass\ transfer}^2 \cdot a_{orb}$. The term $\dot{M}_{cct}$ just determines the amount of mass that is lost from the cct and in mesa is defined as

² Read donut-shaped ring of gas lying in the same plane as the orbit.

```
b% mdot_system_cct = b% mtransfer_rate * b%
↳ mass_transfer_delta
```

and b% mass_tranfer_delta is the fraction of mass lost from circumbinary coplanar toroid.

For now, we assume fully conservative mass transfer, such that $\delta_{mass\ transfer} = 0$ so $\dot{M}_{cct} = 0$. Similarly, since we assume fully conservative mass transfer, $\dot{M}_{transfer} = 0$ for both the donor and the accretor. The accretor is modelled as a point mass and therefore does not lose any mass due to winds.

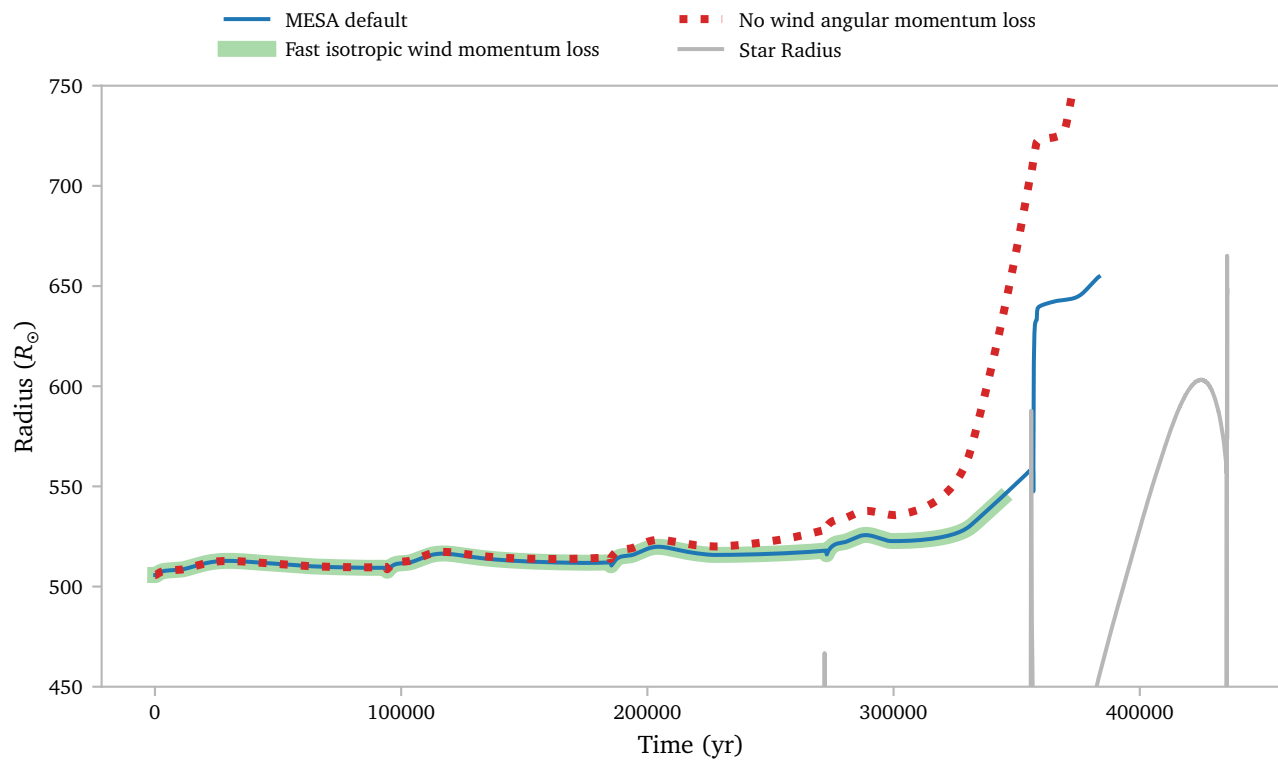
In our case, MESA's implementation is really just

```
subroutine default_jdot_ml(binary_id, ierr)
  integer, intent(in) :: binary_id
  integer, intent(out) :: ierr
  type (binary_info), pointer :: b
  real(dp) :: alfa
  ierr = 0
  call binary_ptr(binary_id, b, ierr)
  if (ierr /= 0) then
    write(*,*) 'failed in binary_ptr'
    return
  end if
  !mass lost from vicinity of donor
  b% jdot_ml = b% mdot_system_wind(b% d_i) *&
    pow2(b% m(b% a_i)/(b% m(b% a_i)+b% m(b% d_i))*b% separation)*2*pi/b% period
end subroutine default_jdot_ml
```

The default MESA implementation of the angular momentum loss due to mass loss can be changed by using the other_jdot_ml hook and writing a custom routine to change b% jdot_ml. To start, I want to test a custom no wind and simple fast wind prescription.

Both are very simple. The first routine simply sets b% jdot_ml = 0 in all cases. This can used as a test. In the case where the angular momentum of the orbit remains unchanged, one would expect the orbits to grow significantly more than in the case where the angular momentum of the orbit *shrinks*. So really, this test should give an upper bound of the orbital radius.

Similarly, I implement the simplified code I wrote above as a routine. This can also be used as a test. It *should* give the same results as the default MESA implementation.



The orbit_evol.py file also implements the fast wind in a very similar way by calculating the change in orbital angular momentum as

```
# Change in semi-major axis, from angular momentum conservation.
dj_dt_orb = (
  (1 - beta)
  * eta
  * a**2
  * np.sqrt(1 - e**2)
  * omega_Kepler(SP.mass + mcomp, a)
  * SP.dM_dt
)
```

In the fast tides mode, $\beta = 0$ and $\eta = (1 + q)^{-2}$ with $q = M_d/M_a$. This thus reduces to

$$\frac{dJ}{dt} = \dot{M}_{d,wind} \left(\frac{M_a a}{M_a + M_d} \right)^2 \omega.$$

This is obviously the same result as the result from MESA. The cool thing however, is that this python file shows how we can modify β and η to use different wind angular momentum loss prescriptions. The most interesting one for us here is the prescription by Saladino et al. (2019).

The code from orbit_evol.py contains the following definitions for β and η .

```
def wind_loss(a, e, spin, mcomp, SP, EvPars):
  ...
  if EvPars.wind_model == "Saladino":
    # Accretion and AM loss from SPH simulations by Saladino et al (2019).
    # (N.B. model results and fit are only for circular orbits...)
    eta = eta_Sal(Q, vw_over_vorb)
    beta = beta_Sal(Q, vw_over_vorb)
  ...

def beta_Sal(q, v):
  """
  Fit to the accretion efficieny from the SPH simulation results of
  Saladino et al (2019) [eq.11 in the paper].
  q = M_loser/M_companion
  v = ratio of outflow velocity to orbital velocity
  """
  c1 = 1.7 + 0.3 * q
  c2 = 0.5 + 0.2 * q
  alpha = 1.0 / (c1 + (c2 * v) ** 5) + 0.75
  beta_max = min(0.3, 1.4 / q**2)
  beta = min(beta_BHL(q, v, alpha), beta_max)
  return beta

def beta_BHL(q, v, alpha=0.75):
  """
  Bondi-Hoyle-Lyttleton accretion efficiency, for a circular orbit.
  q = M_loser/M_companion
  v = ratio of outflow velocity to orbital velocity
  alpha = BHL efficiency parameter (0.75 by default)
  """
  vrel = np.sqrt(1 + v**2)
  beta = alpha / ((1 + q) ** 2 * v * vrel**3)
  return beta

def eta_Sal(q, v):
  """
  Specific orbital angular momentum loss (in units of J/mu = a^2 omega) fitted
  to the SPH simulation results of Saladino et al (2019) [eq.7 in the paper].
  q = M_loser/M_companion
  v = ratio of outflow velocity to orbital velocity
  """
  eta_iso = (1 + q) ** (-2)
  c1 = max(q, 0.6 * q**1.7)
  c2 = 1.5 + 0.3 * q
  eta = min(1.0 / (c1 + (c2 * v) ** 3) + eta_iso, 0.6)
  return eta
```

These functions are all pretty easy to implement in Fortran. For completeness, I wrote them at the end of this document. Below, I show the fortran subroutine that is responsible for the wind angular momentum change.

```

subroutine wind_loss(binary_id, ierr)
  integer, intent(in) :: binary_id
  integer, intent(out) :: ierr
  type(binary_info), pointer :: b
  real(dp) :: q, v
  real(dp) :: beta, eta
  ierr = 0
  call binary_ptr(binary_id, b, ierr)
  if (ierr /= 0) then
    write (*, *) 'failed in binary_ptr'
    return
  end if

  ! --- compute q == m_accretor / m_donor and v
  q = b%m(b%a_i)/b%m(b%d_i)

  ! --- get beta and eta from Saladino
  beta = compute_beta_Sal(q, v)
  eta = compute_eta_Sal(q, v)

  !mass lost from vicinity of donor
  b%jdot_ml = b%mdot_system_wind(b%d_i) &
    *(1 - beta) &
    *eta &
    *(b%separation)**2 &
    *sqrt(1 - pow2(b%eccentricity)) &
    *2*pi/b%period

end subroutine wind_loss

```

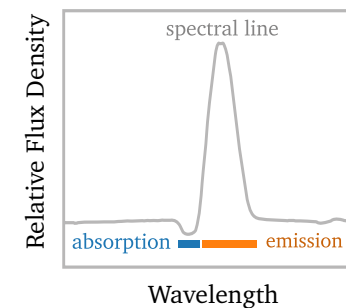
To complete the routine, we need a method to compute the wind velocity. The file `star_model.py` contains a method to compute the wind velocity based on results from Lamers et al. (1995). They determined the terminal velocities v_∞ of the stellar winds of 117 stars using 3 different methods, but most importantly by looking at the violet extend of the black absorption trough of saturated P Cygni profiles. These profiles are special as both an absorption dip, as well as an emission peak can be seen for the same spectral line. This is caused by Doppler shifting of the light, which can be used to derive the escape velocity. I show a schematic depiction in the margin (source: Wikimedia Commons (2024)).

Lamers et al. (1995) find a bistable relation between the escape velocity and the effective temperature of the star shown in figure 4. This relation can quite easily, but not extremely accurately be described by the following set of criteria:

$$v_\infty/v_{\text{esc}} = \begin{cases} 0.35 & \text{if } \log T_{\text{eff}} < 3.5, \\ 1.3 & \text{if } 4.0 \leq \log(T_{\text{eff}}) < 4.35, \\ 2.6 & \text{if } \log(T_{\text{eff}}) \geq 4.35, \\ 0.7 + 0.7(\log T_{\text{eff}} - 4.0) & \text{else.} \end{cases}$$

References

- Jeans, J. H. (1924) *Cosmogonic problems associated with a secular decrease of mass*. [MNRAS85](#), 2.
- Lamers, H. J. G. L. M., Snow, T. P., & Lindholm, D. M. (1995) *Terminal Velocities and the Bistability of Stellar Winds*. [ApJ455](#), 269.
- Saladino, M. I., Pols, O. R., & Abate, C. (2019) *Slowly, slowly in the wind. 3D hydrodynamical simulations of wind mass transfer and angular-momentum loss in AGB binary systems*. [A&A626](#), A68.
- Wikimedia Commons. (2024) *File:P Cygni Profile-en.svg* — *Wikimedia Commons, the free media repository*. [Online; accessed 29-March-2026].



e. Fortran Functions

```
real(dp) function compute_beta_Sal(q, v) result(beta)
  implicit none

  real(dp) :: q, v
  real(dp) :: c1, c2
  real(dp) :: alpha, beta_max

  c1 = 1.7 + 0.3*1/q
  c2 = 0.5 + 0.2*1/q
  alpha = 1.0/(c1 + (c2*v)**5) + 0.75
  beta_max = min(0.3, 1.4*q**2)
  beta = min(compute_beta_BHL(q, v, alpha), beta_max)

end function compute_beta_Sal

real(dp) function compute_beta_BHL(q, v, alpha) result(beta)
  implicit none

  real(dp) :: q, v, alpha
  real(dp) :: vrel

  vrel = sqrt(1 + v**2)
  beta = alpha/((1 + 1/q)**2*v/vrel**3)

end function compute_beta_Sal

real(dp) function compute_eta_Sal(q, v) result(eta)
  implicit none

  real(dp) :: q, v
  real(dp) :: eta_iso, c1, c2

  eta_iso = (1 + 1/q)**(-2)
  c1 = max(1/q, 0.6*(1/q)**1.7)
  c2 = 1.5 + 0.3/q
  eta = min(1.0/(c1 + (c2*v)**3) + eta_iso, 0.6)

end function compute_eta_Sal
```